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Unit 4 : Session 7 : SLO 1 | Generation of Control Signals Fill in the Blanks

I. Fill in the Blanks

1. opcode control signals are generated based on the opcode field of the instruction.
2. control (decoder) unit decodes the instruction to determine which signals to activate.
3. control memory (control store) in microprogrammed control, the control signals are stored as microinstructions in control memory.
4. sequence counter tracks the current step in the instruction cycle.
5. Hardwired hardwired control uses combinational logic to generate signals directly.
6. MEMREAD (memory read) the signal that tells the processor to read from memory.
7. opcode / function-code ALU control signals are determined by the opcode (function field) input to the ALU control.
8. Instruction Register (IR) holds the current instruction being executed.
9. individual bits (one bit per signal) horizontal microprogramming represents each control signal as its own bit in a wide microinstruction word.
10. MEMWRITE (memory write) initiates a write to memory or an I/O device.